

CE 310

Excel Tutorial — Correlation in Excel

Week 7 · Session 2 · Hands-on Walkthrough

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today's Plan — Four New Skills

By the end of this walkthrough you will know how to:

1. Compute the **Pearson correlation coefficient** with `=CORREL(array1, array2)` and identify which variable pair has the strongest relationship
2. Compute r^2 by squaring CORREL and interpret it as explained variance
3. Use **FILTER for conditional CORREL** — filter both arrays to a subgroup before computing r
4. Compute **sample covariance** with `=COVARIANCE.S` and verify it matches $r \times \sigma_1 \times \sigma_2$
5. Build a **scatter chart with trendline** and display R^2 on the chart

Assumed from Weeks 5–6: NORM.DIST, NORM.INV, FILTER, array formulas. Download this week's CSV from D2L before you start.

Step 1 — Confirm the Dataset

File: `Week5_D0E_MedOffice_ZoneTemp_2023.csv` · download from D2L · 8,760 rows

Column	Header	Type
G	Zone_Temp_F	Number
H	Outdoor_Temp_F	Number
I	Cooling_kWh	Number
F	BusinessHours	Text ("Yes"/"No")

 In a blank cell, type `=COUNTA(A2:A8761)` → should return **8760**.

Open **Wk07.04_In_Class_Answer_Sheet.xlsx** — enter your name. You will compute CORREL on columns G, H, and I. Today we use all three for the first time.

Step 2 — CORREL: Syntax Before You Type

Two arguments, both arrays, must be the same length. `=CORREL(array1, array2)` — array1 is the first variable, array2 is the second. Both must cover the same rows. `=CORREL(G2:G8761, H2:H8761)` computes $r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$. Order does not matter: $r(X,Y) = r(Y,X)$.

CORREL and PEARSON give identical results. Excel has both functions. This course uses CORREL. If you see PEARSON in documentation, it computes the same number.

 Type these in blank cells and verify:

`=CORREL(G2:G8761, H2:H8761)` → $r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ → expected: **0.7529**

Press Enter — you should see a number near 0.75. This is a moderate-to-strong positive relationship.

Interpretation: when outdoor temperature rises, zone temperature tends to rise too — but the HVAC limits how much. In this dataset: $\sigma_{\text{outdoor}} = 16.9^{\circ}\text{F}$, $\sigma_{\text{zone}} = 2.9^{\circ}\text{F}$ (5.8× compression).

Step 3 — Three Pairs, Spot the Weak One

 Type and verify — enter all three in blank cells:

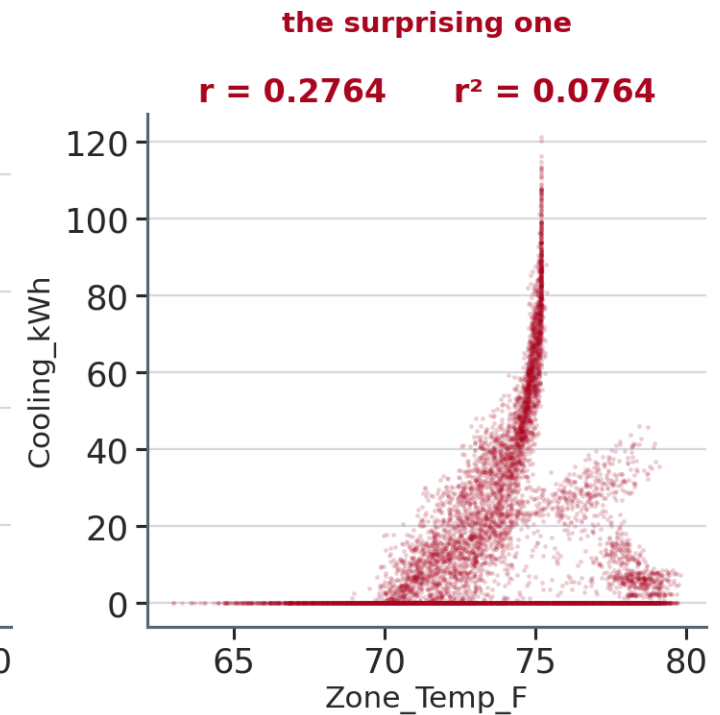
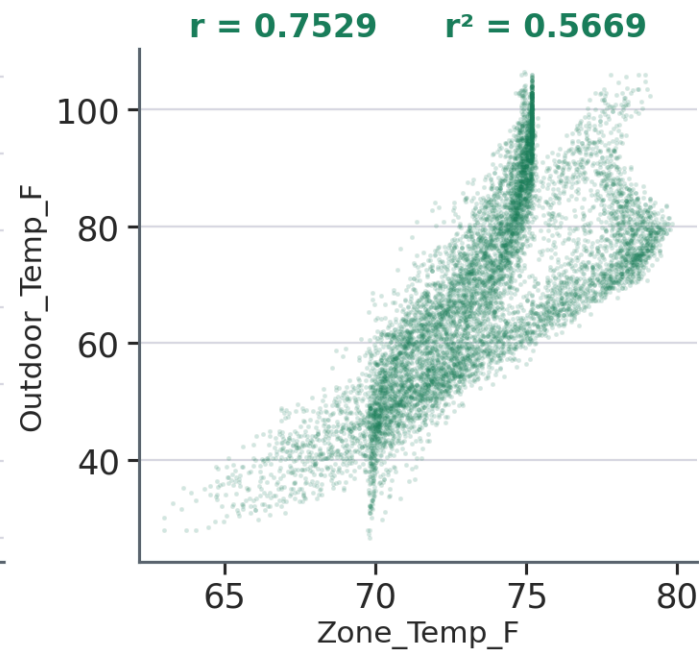
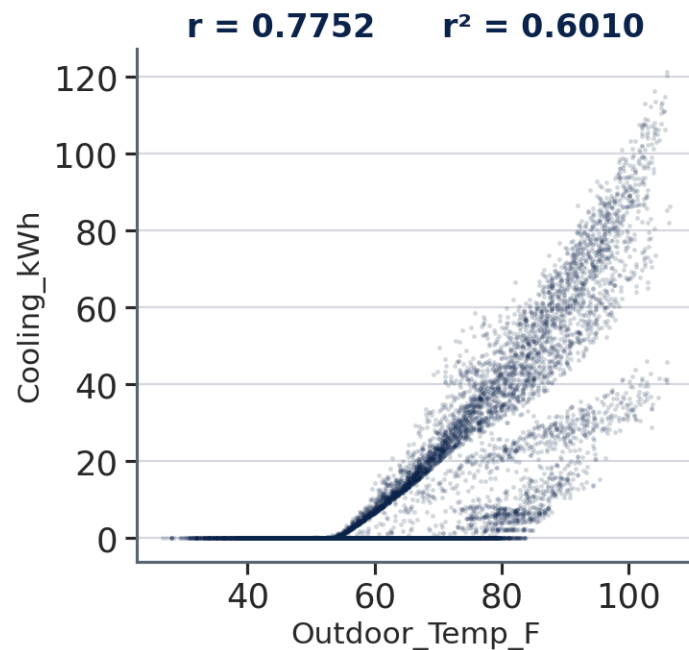
`=CORREL(H2:H8761, I2:I8761)` → r(Outdoor, Cooling) → expected: **0.7752** ← the main pair

`=CORREL(G2:G8761, I2:I8761)` → r(Zone, Cooling) → expected: **0.2764** ← the surprising one

Pair	r	Strength
Outdoor_Temp_F vs Cooling_kWh	0.7752	Strong positive
Zone_Temp_F vs Outdoor_Temp_F	0.7529	Moderate-strong positive
Zone_Temp_F vs Cooling_kWh	0.2764	Weak — surprising

Step 3 – The Three Pairs, Plotted

Same 8,760 hours, three pairs of columns

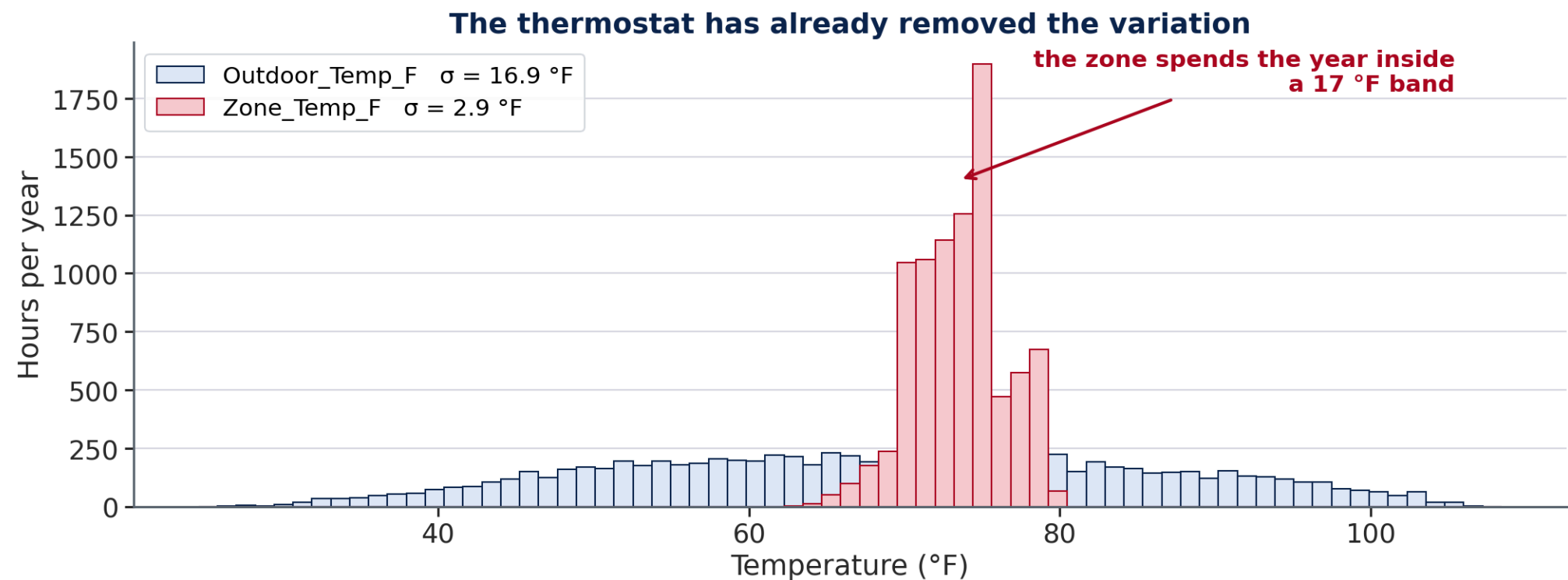


Step 3 (continued) — Why Is $r(\text{Zone}, \text{Cooling})$ Weak?

Many students expect $r(\text{Zone}, \text{Cooling})$ to be HIGH — "if the zone is hot, cooling must be high." The direction is correct (positive), but the magnitude is weak.

⚠ Zone temperature is tightly controlled near the setpoint ($\sigma_{\text{zone}} \approx 2.9^\circ\text{F}$) — it barely varies. When a variable barely varies, it cannot carry a strong correlation with anything else. Outdoor temperature ($\sigma = 16.9^\circ\text{F}$, no thermostat) carries the signal.

Step 3 – A Variable That Barely Moves



Step 4 — r^2 by Squaring

No new Excel function — just squaring the result.

r^2 is the coefficient of determination — the fraction of one variable's variance linearly explained by the other.

 In blank cells:

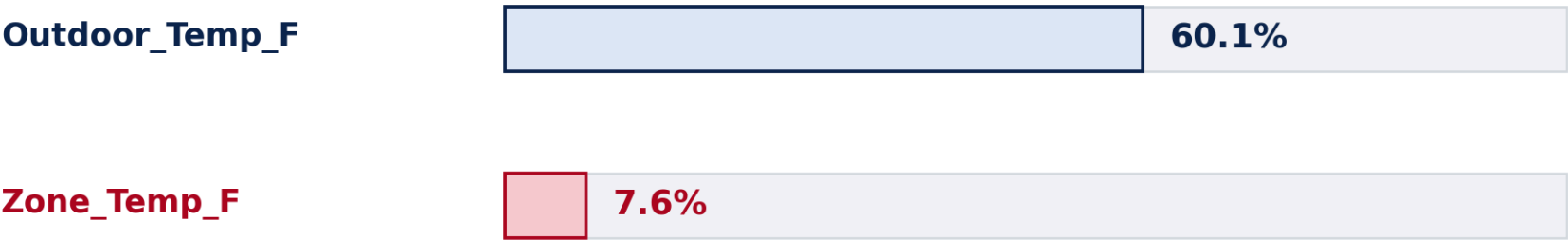
= (your CORREL cell)^2 → or type =CORREL(H2:H8761, I2:I8761)^2 → expected: **0.6010**

Interpretation: outdoor temperature explains **60.1%** of cooling variance across all 8,760 hours. The remaining **39.9%** comes from solar gain, occupancy differences, HVAC scheduling, and other factors outdoor temperature alone cannot capture.

For the weak pair: =CORREL(G2:G8761, I2:I8761)^2 → **0.0764** — zone temperature explains only 7.6% of cooling variance. Outdoor temperature is roughly **8x more informative** (60.1% vs 7.6%).

Step 4 — What 60.1 % Actually Claims

r^2 is the share of cooling variance the column explains



Outdoor temperature is about 8× more informative about cooling load than the zone temperature is.

Step 5 — FILTER for Conditional CORREL

To compute r for a subgroup, you must filter **both** arrays — not just one. If you filter only the first array, the two arrays have different lengths and CORREL returns an error. Wrap each array in FILTER with the same condition.

Pattern: `=CORREL(FILTER(array1, condition), FILTER(array2, condition))`

 Business hours (F="Yes"):

`=CORREL(FILTER(H2:H8761, F2:F8761="Yes"), FILTER(I2:I8761, F2:F8761="Yes"))`

→ $r(\text{Outdoor, Cooling})$ biz hours → expected: **0.9539**

 Summer only (E="Summer"):

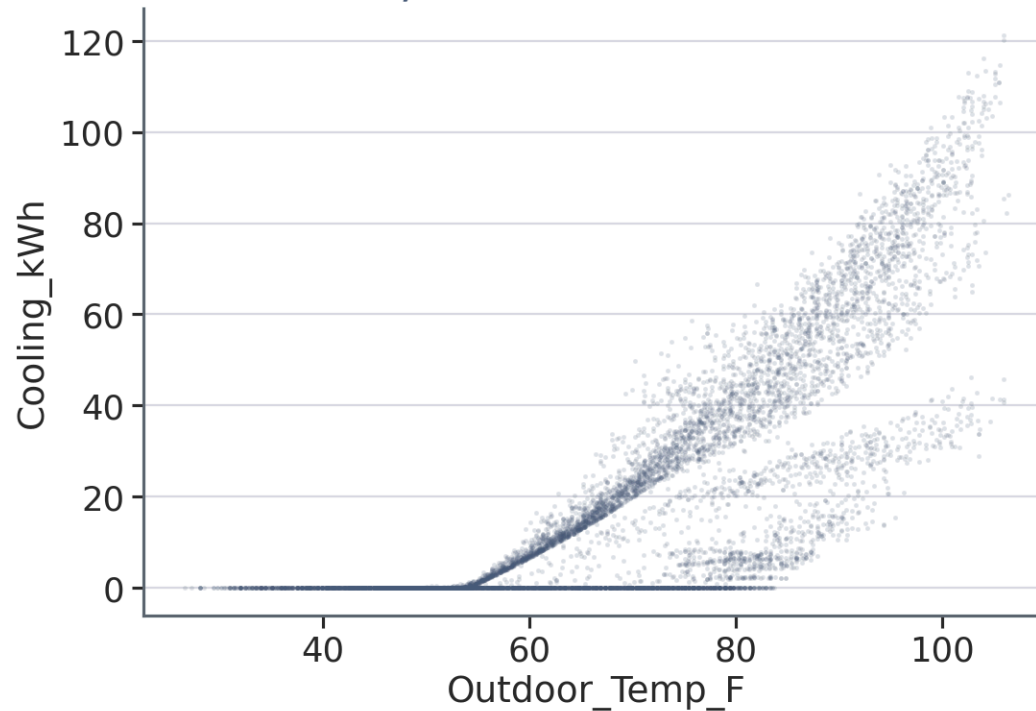
`=CORREL(FILTER(H2:H8761, E2:E8761="Summer"), FILTER(I2:I8761, E2:E8761="Summer"))`

→ $r(\text{Outdoor, Cooling})$ Summer → expected: **0.7494**

Key result: $r = 0.9539$ during business hours vs. 0.7752 for all hours.

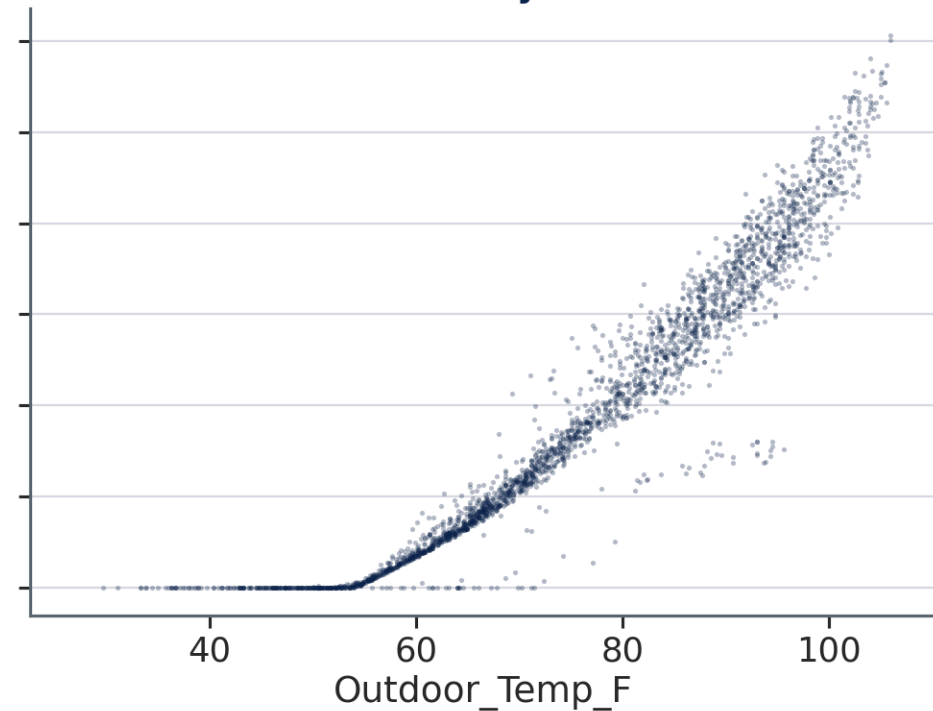
Step 5 — What Filtering Does to r

Filter BOTH arrays — one filtered array is a length mismatch
All 8,760 hours $r = 0.7752$



nights and weekends sit along the floor —
the plant is off whatever the weather does

Business hours only $r = 0.9539$



remove them and r rises from 0.78 to 0.95

Step 6 — Array Formula for Older Excel

⚠ FILTER requires Excel 365 or Excel 2019+. If FILTER shows #NAME?, use the array formula version below.

The pattern uses IF instead of FILTER:

```
{=CORREL(IF(condition, array1), IF(condition, array2))}
```

Press **Ctrl+Shift+Enter** (not just Enter). Excel adds {} braces automatically — do NOT type them manually.

📌 Same result as Step 5, array formula version:

Business hours:

```
{=CORREL(IF(F2:F8761="Yes", H2:H8761), IF(F2:F8761="Yes", I2:I8761))}
```

→ **0.9539**

Summer:

```
{=CORREL(IF(E2:E8761="Summer", H2:H8761), IF(E2:E8761="Summer", I2:I8761))}
```

→ **0.7494**

Step 7 — COVARIANCE.S

COVARIANCE.S uses $n-1$ in the denominator (same convention as STDEV — sample data).

COVARIANCE.P uses n — correct only for a complete population census. Use COVARIANCE.S for sample data. The difference is tiny for $n=8,760$, but the approach is wrong with COVARIANCE.P.

 Type and verify:

=COVARIANCE.S(H2:H8761, I2:I8761) → cov(Outdoor, Cooling) → expected: **331.58 °F·kWh**

=COVARIANCE.S(G2:G8761, H2:H8761) → cov(Zone, Outdoor) → expected: **37.15 °F²**

Cross-check: $r = \text{cov}(X,Y) / (\sigma_X \times \sigma_Y)$. Verify your covariance:

$331.58 / (16.918 \times 25.282) = 331.58 / 427.6 = \mathbf{0.7752} \checkmark$ matches your CORREL result.

$37.15 / (2.916 \times 16.918) = 37.15 / 49.3 = \mathbf{0.7529} \checkmark$ matches your CORREL result.

Step 8 — Scatter Chart with Trendline

 Select **H1:I8761** (columns H and I, including headers)
→ Insert → Charts → Insert Scatter (X,Y) → Scatter (dots only)
→ Chart appears: Outdoor_Temp_F on X-axis, Cooling_kWh on Y-axis
→ Right-click any data point → **Add Trendline**
→ Linear → check "*Display R-squared value on chart*" → Close
→ Double-click chart title to rename: "*Outdoor Temperature vs. Cooling Load — All Hours*"

Verify the R^2 on the chart: $R^2 \approx 0.60$ — matches your CORREL^2 result (0.6010). If the chart R^2 is very different, check that you selected H and I columns (not G and I or G and H).

Common mistakes:

- Line chart instead of scatter: right-click series → Change Chart Type → Scatter
- Dots connected by lines: choose "Scatter (X,Y)" NOT "Line"
- R^2 missing: right-click trendline → Format Trendline → check "Display R-squared"

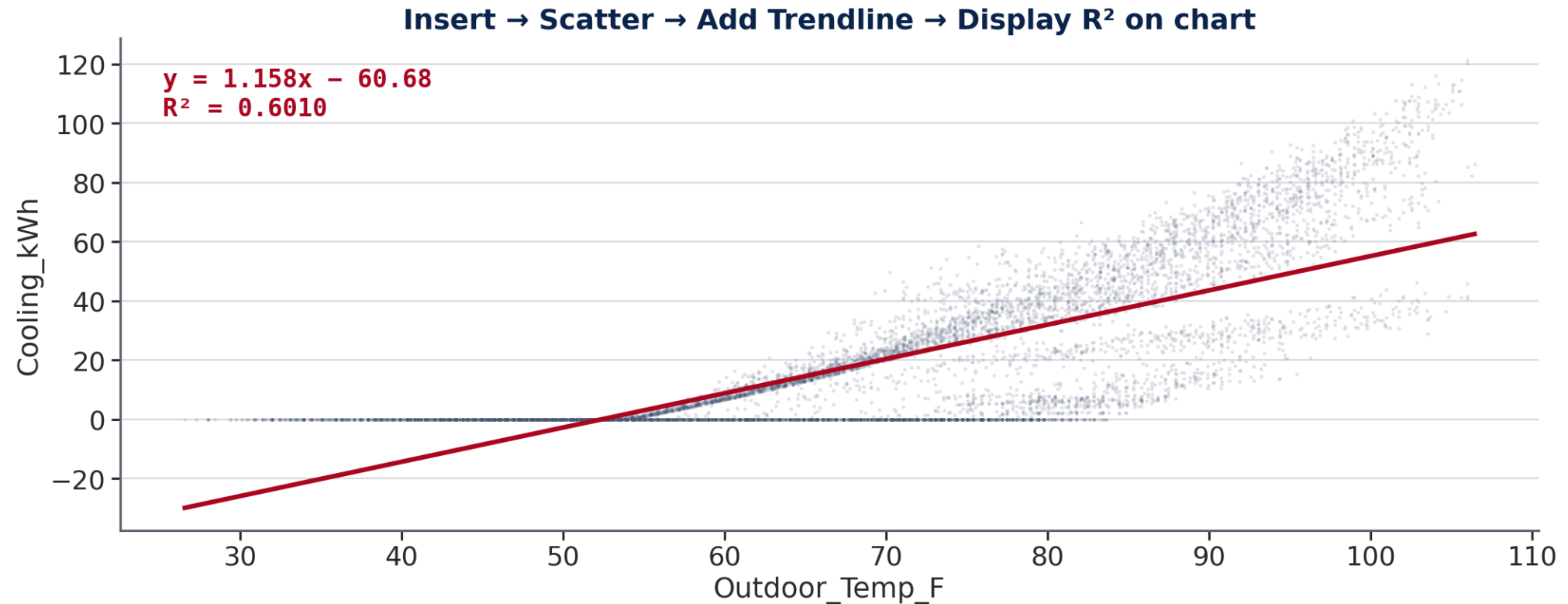
Step 9 — Chart 2: Two Series Need a Legend

Chart 2 plots the same relationship **split by season** — two series on one scatter, so a reader needs to know which is which.

 **Chart Design → Add Chart Element → Legend**

Excel usually adds one as soon as there is a second series, so check before adding a second. Same menu as the axis titles and the trendline.

Step 8 — What Your Chart Should Look Like



Quick Check — Before You Start the In-Class Exercise

Think, then check

You computed:

- $r(\text{Outdoor, Cooling}) = 0.7752 \cdot r^2 = 0.6010$
- $r(\text{Zone, Cooling}) = 0.2764$
- $r(\text{Outdoor, Cooling}) \text{ biz hours} = 0.9539$

Question A: Write the FILTER formula for $r(\text{Outdoor, Cooling})$ restricted to Winter only. What do you expect — higher or lower than the all-hours 0.7752?

Question B: If $\text{COVARIANCE.S}(H:H, I:I) = 331.58$, $\text{STDEV}(H:H) = 16.918$, and $\text{STDEV}(I:I) = 25.282$, what is $r(\text{Outdoor, Cooling})$? Compute it from these three numbers.

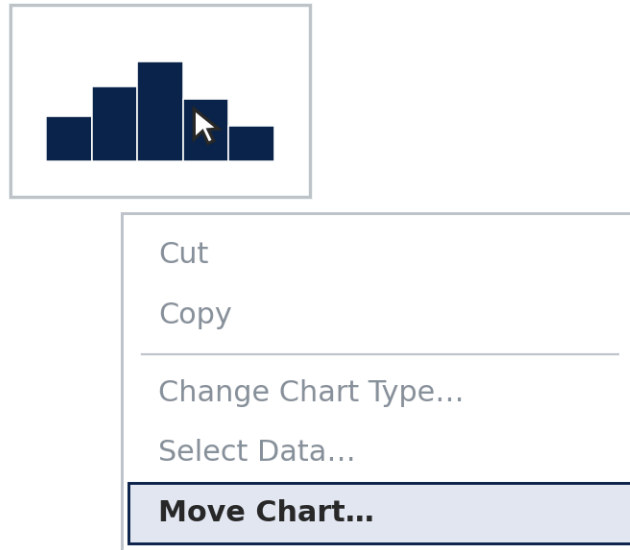
Compare your answers with your neighbor.

The Step Every In-Class Exercise Ends With

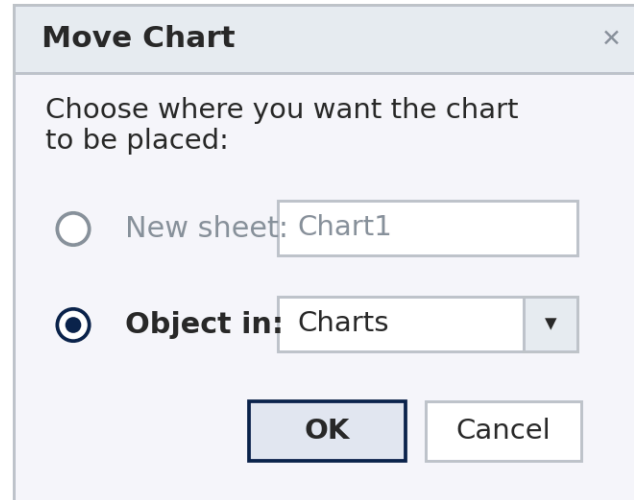
Getting a finished chart into the box it is graded in

Every week. Build the chart on the sheet its data lives on, then move it onto the Charts sheet.

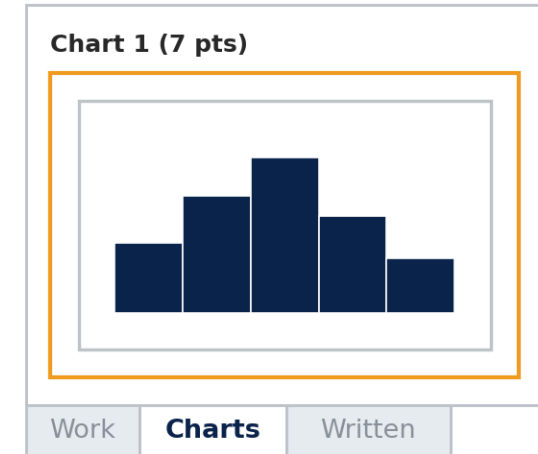
1 Right-click the chart



2 Choose "Object in"



3 It lands in the box



✗ New sheet — makes a separate sheet called Chart1. The box on Charts stays empty and the points are lost.

✓ Object in → Charts — then drag the chart inside the outlined box.

⚠ Leave it a live chart — a pasted picture of one scores zero.

Quick Reference — Week 7 New Formulas

Task	Formula
Correlation	<code>=CORREL(col1, col2)</code>
Coefficient of determination	<code>=CORREL(col1, col2)^2</code>
Conditional r (FILTER)	<code>=CORREL(FILTER(col1, cond), FILTER(col2, cond))</code>
Conditional r (array)	<code>{=CORREL(IF(cond, col1), IF(cond, col2))}</code>
Sample covariance	<code>=COVARIANCE.S(col1, col2)</code>

Pair	r	r ²
Outdoor vs Cooling (all)	0.7752	0.6010
Zone vs Outdoor (all)	0.7529	0.5669
Zone vs Cooling (all)	0.2764	0.0764
Outdoor vs Cooling (biz)	0.9539	0.9100
cov(Outdoor, Cooling)	331.58 °F·kWh	—

Older Excel: use Ctrl+Shift+Enter.

Before You Submit

⚠ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

1. **File format** — **.xlsx only**. Save → Save As → Excel Workbook (*.xlsx). Files saved as **.xls**, **.csv**, or **.pdf** cannot be graded.
2. **Sheet names** — **Answers** and **Written**. Do not rename or delete either sheet in your workbook. The assignment has its own file, **Wk07.06_Assignment_Answer_Sheet.xlsx**.
3. **Column layout** — **do not change**. Answers go in Column C. Do not insert, delete, or move columns.
4. **File name** — **CE310_W7_<NetID>.xlsx** (replace **<NetID>** with your UA NetID).
5. **Upload** the **.xlsx** file to D2L Dropbox by **9:00 AM next Wednesday**.

You're Ready — Open the In-Class Exercise Instructions

[Wk07.03_In_Class_Exercise.pdf](#) · [Wk07.04_In_Class_Answer_Sheet.xlsx](#)

Work independently. Raise your hand if you get stuck.